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COMPENSATING SET SCREW FOR TERMINAL BLOCKS TO COMPENSATE FOR SETTLING OF MULTI-STRAND WIRE

Abstract

A computing system includes one or more terminal blocks for terminating power wires, which may be used for delivering power to or distributing power within the computing system. A set screw is provided for each terminal block, with the set screw configured to hold a wire in the terminal block. At least one of the set screws is a compensating set screw. The compensating set screw has a set screw head, a set screw stem, a plunger movable along the stem, a spring mechanism between the head and the plunger, and a retainer to retain the plunger on the stem. The spring mechanism may push the plunger toward the wire, causing the plunger to follow the wire and maintaining pressure between the plunger and the power wire even as the power wire settles over time.

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Background/Summary

INTRODUCTION

[0001] Large enterprise-grade computer systems can require a great amount of power. In particular, computer systems which comprise multiple compute units (e.g., “blades” or “trays”) aggregated together, such as High Performance Compute (HPC) systems, may require thousands or tens-of-thousands of watts of power. Power is often delivered to these large systems and/or distributed within these systems by multi-strand power wires (a multi-strand wire comprises many smaller conductive strands which are bundled together to form a single conductor, as opposed to single-strand (or “solid”) power wires in which each individual conductor comprises a single monolithic wire). For example, a computer system with multiple compute units may have a power distribution unit (PDU) which receives input power from an external power source (e.g., a utility) via a first set of conductors and distributes the power to the individual compute units in the system via a second set of conductors, with the first set and/or the second set of conductors comprising multi-strand wires. Often, the multi-strand wires may be hardwired to terminal blocks, as opposed to being removably connected via a plug or connector. For example, the multi-strand power lines may enter the PDU of a computer system and be terminated at a terminal block of the PDU, thereby electrically connecting the power wire to internal electrical circuitry in the PDU. In addition, there may be various similar terminal blocks within the PDU or elsewhere in the computing system to establish electrical connections to other wires which aid in distributing the power to the various components of the large computer systems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] The present disclosure can be understood from the following detailed description, either alone or together with the accompanying drawings. The drawings are included to provide a further understanding of the present disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate one or more examples of the present teachings and together with the description explain certain principles and operation. In the drawings:

[0003] FIG. 1 is a block diagram illustrating an example computer system including an example electrical termination including a terminal block and compensating set screw.

[0004] FIG. 2 is perspective view of an example compensating set screw.

[0005] FIG. 3 is another perspective view of the compensating set screw of FIG. 2.

[0006] FIG. 4 is another perspective view of the compensating set screw of FIG. 2.

[0007] FIGS. 5A-B are exploded perspective views of the compensating set screw of FIG. 2. FIG. 5A shows the set screw from a bottom perspective. FIG. 5B shows the set screw from a top perspective.

[0008] FIGS. 6A-6D comprise schematic diagrams illustrating example configurations of spring mechanisms of a compensating set screw. FIG. 6A illustrates a first configuration in which the spring mechanism comprises a single spring. FIG. 6B illustrates a second configuration in which the spring mechanism comprises multiple springs arranged in series with the same orientations. FIG. 6C illustrates a third configuration in which the spring mechanism comprising multiple springs arranged with alternating orientations. FIG. 6D illustrates a fourth configuration in which the spring mechanism comprises multiple groups of springs, with each group having multiple springs with the same orientation as one another, and with the groups having alternating orientations relative to one another.

[0009] FIGS. 7A-B are cross-sectional views of an example electrical termination comprising a terminal block and the compensating set screw of FIG. 2. FIG. 7A illustrates the electrical termination in a first state in which the set screw is compressed. FIG. 7B illustrates the electrical termination in a second state in which the set screw is partially decompressed.

[0010] FIGS. 8A-B are cross-sectional views of an example electrical termination comprising a terminal block and an example set screw. FIG. 8A illustrates the electrical termination in a first state in which the set screw is compressed. FIG. 8B illustrates the electrical termination in a second state in which the set screw is partially decompressed

[0011] FIG. 9 is a perspective view of an example terminal block unit.

DETAILED DESCRIPTION

[0012] As noted above, some computing systems (particularly large-scale computing systems) may include terminal blocks to terminate electrical wires which supply power to and/or distribute power within the system. At each terminal block, there are one or more set screws used to hold the power wires in place. Each wire is inserted into an opening in the terminal block, and a corresponding set screw is screwed into the terminal block and presses down on the power wire to hold it in place. When the wire terminated at the terminal block is a multi-strand wire, over time, the power wire may settle due to conditions such as heat fluctuations and vibration. This is especially prevalent in high strand count power wires often used in larger computing systems, as they tend to settle more than low strand count wires. High strand count wire is more flexible than lower strand count wire. As space in large-scale computing systems gets smaller, higher strand count wire becomes more likely to be used to distribute the power to and within the large-scale computing systems.

[0013] As the power wires settle, the contact pressure between the power wires and the set screw reduces and, in some cases, the wire may become loose in the terminal block. This causes the electrical resistance at the interface between the wire and the terminal block to increase. The increase in resistance leads to increased heat and can even cause fires. Adding a second set screw may sometimes help hold the power wire in place, but since the power wire is likely to settle under both set screws, it cannot solve the problem. Due to the risk of fires from this, maintenance tasks are required to re-torque the set screws in all of the terminal blocks. The process may require the PDU and other locations of the terminal blocks to be powered off during the maintenance. This can be very time-consuming and expensive for the operator of the computing system due to the required downtime. It also requires a design where the terminal blocks must be readily available to be re-torqued. This further restrains the design of the PDU and other locations.

[0014] To address these and other issues, examples disclosed herein include computing system terminal blocks which have compensating set screws that can maintain suitable contact pressure with multi-strand power wires as the power wires settle. These set screws each comprise a spring-loaded plunger which can move along a stem of the screw. When the set screw is initially tightened, a spring mechanism of the set screw is compressed, resulting in spring forces which press the plunger against the wire. Thus, as the wire settles, the spring mechanism causes the plunger to follow the wire and to continue to press against the wire. In this manner, high contact pressure can be maintained with the wire notwithstanding its settling, thus avoiding an increase in resistance due to loosening of the wires.

[0015] An example compensating set screw may include a threaded set screw head, a set screw stem, a set screw spring mechanism, a plunger and a plunger retainer. The set screw head may be configured with external threading surrounding the exterior and a tool receptacle on top of the set screw head where the appropriate tool can be used to tighten or loosen the set screw within the terminal block. The set screw stem is attached to the set screw head and extends below the set screw head. The spring mechanism is placed on the set screw stem below the set screw head. The plunger is moveably attached to the set screw stem below the spring mechanism. Below that, on the set screw stem is the retainer, which is configured to hold the plunger moveably attached to the set screw stem.

[0016] Because the spring mechanism is positioned between the head of the set screw and the plunger, the position of the plunger along the stem determines a state of compression of the spring mechanism—i.e., the closer the plunger is to the head, the more compressed the spring mechanism. For example, the spring mechanism becomes loaded (compressed) when the set screw is screwed

into the terminal block, as the tightening of the screw and contact between the plunger and the wire cause the plunger to move along the stem to a retracted position (the retracted position being nearer to the head than an initial position of the plunger). The compression of the spring mechanism results in a restoring spring force pushing the plunger away from the head and against the power wire. The power wire is thus initially secured by the plunger in this retracted position. As the power wire settles, the spring mechanism urges the plunger of the set screw to follow the power wire and move along the set screw stem towards an extended position. In this extended position of the plunger, the spring mechanism continues to generate spring forces which push the plunger against the wires and maintain contact pressure on the power wire.

[0017] In some examples, the spring mechanism may include one or more disk springs. Belleville washers could be used, for example. A single disk spring could be used, or stacks of disk springs could provide even more pressure against the power wire. The stacks could be configured as a stack of parallel disk springs. They could also be configured in pairs of opposing disk springs. Since the disk springs are conical, when stacked in a pair in opposite directions, they provide more range of motion than would two disk springs stacked in parallel. This pair of opposing disk springs may be repeated to further increase the range of motion. These two configurations may be combined so that the spring mechanism consists of a stack of disk springs in parallel with each other stacked on top of a second stack of disk springs in parallel with each other where the two stacks are aligned in opposite directions. Stacks may also be formed of two or more disk springs in any pattern of upright and inverted disk springs. These different configurations of the spring mechanism allow for compensating set screws with many different configurations of force and range of motion. Other spring mechanisms are possible, including the use of one or more helical (spiral) compression springs.

[0018] Terminal blocks are typically rated to a set torque value that corresponds to a certain amount of clamping force. They need to be able to apply a set amount of clamping force initially as well as after the settling of the power wire. Having a highly configurable compensating set screw allows for maintaining sufficient force in many different situations with different types of power wire. This is important to ensure that sufficient clamping force is still applied after the power wire settles, regardless of the strand count of the wire being used.

[0019] Terminal blocks configured with these set screws may be deployed at multiple locations in the large computer system, and specifically in multiple parts of the power distribution unit (PDU) in some examples. The terminal block is configured with a threaded set screw hole for accepting the set screw by meshing with the threads of the set screw as the set screw is screwed into the terminal block. The terminal block also includes a wire receptacle for accepting the power wire. The terminal block is configured such that the set screw is screwed into the terminal block with an appropriate tool, pushing the plunger against the power wire in the wire receptacle such that the power wire is held in place.

[0020] In some examples, the terminal block of the PDU that first receives the power from a power source into the computing system is located at the input unit of the PDU. It is most likely to see high strand count wires and therefore most likely to experience the settling of the power wires. However, within the PDU, other terminal blocks can experience this problem and thus would benefit from terminal blocks configured with the disclosed set screw. For example, PDUs often include circuit breaker mechanisms, such as fuses, within the PDU itself. As the power is connected into and out of the circuit breaker mechanism it is often by way of a terminal block. Additionally, power connections coming out of the output unit of the PDU to be distributed to the individual compute trays and other components of the large computer system will go through terminal blocks.

[0021] In some computing systems, multiple compute trays are aggregated together, and each individual compute tray of the large computer system will typically include its own power supply unit (PSU). The PSU may be configured to convert AC power delivered from the PDU to DC power for use in the compute tray. The PSU may also be configured to convert one voltage level to

another. The power exiting the PDU is distributed to the multiple compute trays where it is received by the PSUs. In some examples, one of the disclosed terminal blocks may be used at the PSU to receive the power from the PDU.

[0022] Each of the compute trays includes a chassis configured to hold the individual components of the compute tray, such as the PSU and system board. The system board includes one or more processors. The multiple compute trays are often included as part of a much larger computer system held in a rack system together with other components of the large computer system, such as the PDD, cooling infrastructure, networking systems, and storage systems.

[0023] These and other examples will be described in greater detail below in relation to FIGS. 1-9.

[0024] FIG. 1 illustrates an example electrical termination **105** and an example computer system **100** comprising the same. The electrical termination **105** is illustrated in an installed state in the computer system **100** to aid understanding, but in some examples the electrical termination **105** could be provided (e.g., manufactured, sold, etc.) separately from the other components of computer system **100**. In addition, the electrical termination **105** is illustrated as a part of a power distribution unit (PDU) to aid understanding, but in various examples the electrical termination **105** could be provided elsewhere in the computing system **100**, in addition to or in lieu of being provided as part of the PDU. In particular, the electrical termination **105** may be provided anywhere in the computing system **100** where it is desired to terminate a power wire.

[0025] The computer system **100** may comprise one or more Power Distribution Units (PDUs) **110** and one or more compute trays **160**. While FIG. 1 only shows one of each, it should be understood that more than one PDU **110** and more than one compute tray **160** is possible within the computer system **100**.

[0026] The compute tray **160** includes a chassis **161**, power supply unit **162** and system board **163**. The chassis **161** may be configured to hold the system board **163** and the power supply unit **162**. The system board **163** may be configured to hold one or more processors **164**. The power supply unit **162** is configured to receive power from the PDU **110** and provide it to the other components of the compute tray **160**, such as the system board **163**. The power supply unit **162** may include one or more terminal blocks **130**.

[0027] The PDU **110** has an input unit **120** and an output unit **140**. The input unit **120** is configured to receive power from an external power source, such as a utility power source, and may comprise electrical terminations to terminate (electrically connect to) external wires which carry the power from the external power source. The output unit **140** is configured to supply the power to the power supply units **162** of compute trays **160** and may comprise electrical terminations to electrically connect to wires or other conductors (e.g., bus bars) which connect the PDU **110** to the power supply units **162**. In some examples, one or more of the electrical terminations of the input unit **120** and/or the output unit **140** comprises an electrical termination **105**. The electrical terminations **105** each comprise a terminal block **130** and at least one compensating set screw **170**.

[0028] In an example, the input unit **120** includes at least one electrical termination **105** comprising a terminal block **130** configured to terminate an external power wire from an external power source. Thus, in such examples, the terminal block **130** of input unit **120** is where the input power first enters the computing system **100**. From the input unit **120**, the power is transferred through internal circuitry (e.g., internal power wires, bus bars, or other conductors) to other components of the PDU **110**. In some examples, one of the other components of the PDU **110** may be a circuit breaker mechanism **150**. While only one circuit breaker mechanism **150** is shown, it should be understood that more than one may be present in the PDU **110**. The circuit breaker mechanism **150** may be a fuse, for example. In some examples, the circuit breaker mechanism comprises one or more of the electrical termination **105** that terminate internal PDU power wires, such as internal power wires coming from the input unit **120** and/or internal power wires going to the output unit **140**. Power coming out of the circuit breaker mechanism **150** may be routed to other components and is eventually routed to the output unit **140**. As noted above, in some examples, the output unit

140 may include one or more electrical terminations **105** comprising terminal block **130**. From the output unit **140**, power is routed from the PDU **110** to other components of the computer system **100**, such as the compute tray **160**.

[0029] As noted above, each electrical termination comprises a terminal block **130**. The terminal block **130** may be configured with one or more set screw holes **131** and corresponding wire receptacles **132** (e.g., at least one set screw hole **131** per wire receptacle). Typically, there will be one set screw hole **131** per wire receptacle **132** but it is possible to configure more than one set screw hole **131** to a wire receptacle **132**. Each set screw hole **131** is threaded and configured to receive a set screw **170** by screwing the set screw **170** into the matching threads of the set screw hole **131**. Thus, in some instances, each electrical termination **105** may comprise the same number of set screws **170** as there are set screw holes **131** in the terminal block **130**. The set screw hole **131** extends through the terminal block **130** and intersects (joins) with the corresponding wire receptacle **132** in such a way that, when the set screw **170** is screwed into the terminal block **130**, it extends into the wire receptacle **132** to press against the power wire, holding it in place.

[0030] The set screw **170** includes a set screw head **171** and a set screw stem **175**. The set screw head **171** includes a tool receptacle **172** on one end where the appropriate tool can be used to tighten or loosen the set screw **170** within the terminal block **130**. Around the outer edge of the set screw head **171** is configured with threading **173** to mate with the threading on the set screw hole **131** in order to screw the set screw **170** into the terminal block **130**. The set screw stem **175** extends below the set screw head **171**. In some examples, the set screw stem **175** is integrally connected to (i.e., formed as part of the same monolithic body as) the set screw head **171**. In other examples, the set screw stem **175** is formed separately from the set screw head **171** and is later attached thereto, e.g., by mechanical fasteners (e.g., external threads on the stem **175** which mate with internal threads in a hole in a bottom side of the head **171**), welding, or other fastening techniques.

[0031] Towards the distal end portion of the set screw stem **175**, opposite the set screw head **171**, is attached a retainer **176**. The retainer **176** is configured to hold the plunger **177** moveably attached to the set screw stem **175** (i.e., the retainer **176** blocks the plunger **177** and prevents it from coming off the distal end of the stem **175**). In some examples, the retainer **176** may include a retaining clip, retaining washer, or similar component capable of holding the plunger **177** on the set screw stem **175** and allowing the plunger **177** to move up and down along the set screw stem **175**.

[0032] The plunger **177** is configured with an opening for the set screw stem **175**. There is a small plunger hole **178** on the portion of the plunger **177** closest to the spring mechanism **174** and a large plunger hole **179** on the opposite portion of the plunger **177**. The small plunger hole **178** is smaller than the size of the retainer **176**. The large plunger hole **179** is larger than the size of the retainer **176**. This configuration allows the plunger **177** to move up and down along the set screw stem **175** while the small plunger hole **178** prevents the retainer **176** from passing through thus holding the plunger **177** moveably attached to the set screw stem **175**.

[0033] In some examples, the retainer **176** is formed separately from the stem **175** and is later attached to the stem **175**. For example, the retainer **176** may be removably attached to the stem, such as by a snap-fit or friction fit, or may be permanently attached to the stem **175**, such as by welding. In other examples, the retainer **176** is integrally connected to (formed as part of the same monolithic body as) the stem **175**. For example, the retainer **176** may be formed from a portion of the stem **175** which has been bent, deformed (e.g., like a rivet), or otherwise shaped so as to prevent the plunger **177** from passing of the distal end of the stem **175**.

[0034] In some examples, the retainer **176** is attached to or formed in the stem **175** after the spring mechanism **174** and plunger **177** have been assembled onto the stem **175**. This approach may be particularly useful in those examples in which the stem **175** is formed integrally with the set screw head **171**, as it may allow for the spring mechanism **174** and plunger **177** to be assembled onto the stem **175** by passing them over a distal end of the stem **175** prior to attachment or formation of the retainer **176**. In other examples, the retainer **176** is attached to or formed in the stem **175** prior to

the spring mechanism **174** and plunger **177** being assembled onto the stem **175**. This approach may be usable when the stem **175** is formed separately from the head **171**, as this may allow the spring mechanism **174** and plunger **177** to pass over the proximal end of the stem **175** prior to the stem **175** being attached to the head **171**.

[0035] Situated between the set screw head **171** and the plunger **177**, along the set screw stem **175** is the spring mechanism **174**. The spring mechanism **174** includes one or more springs. In one example, the springs may be disk springs. In other examples, the springs may be helical (spiral) compression springs.

[0036] In examples in which the springs are disk springs, each disk spring may have an outer edge, a central opening, and an inner edge defining the central opening. Moreover, in a resting (uncompressed) state, the inner edge is offset from the outer edge along an axis for the central opening. In other words, rather than being flat disks, the disk springs have the shape of a truncated hollow cone with a central bore, such that the inner edge protrudes above or below the outer edge. In these examples, the spring mechanism **174** can include many different configurations of disk springs. In one configuration, there is a single disk spring. In a second configuration, there are two or more disk springs stacked in series, with all of the disk springs having the same orientation. (Note that, as used herein, the orientation of the disk springs is defined by the direction that the inner edge thereof protruded relative to the outer edge; for example, in a first orientation the inner edge protrudes upward and in a second orientation the inner edge protrudes downward.) In a third configuration, one or more pairs of disk springs are stacked, where each pair consists of two adjacent disk springs having opposite orientations, i.e., stacked in opposite directions from each other. In other words, the disk springs alternate in their orientations. In a fourth configuration, one more sets of disk springs are provided stacked, with each set of disk springs comprising two subsets: a first subset with two or more disk springs stacked in series and having the same orientation as one another, and a second subset, stacked on the first subset, and comprising two or more disk springs stacked in series and having the same orientation as one another but a different orientation than the first subset. For example, each set could comprise four disk springs with a first subset of two disk springs with a first orientation stacked on a second set of two more disk springs having a second orientation. The spring mechanism **174** may consist of any number of disk springs stacked in any configuration of upright and opposite configurations.

[0037] The set screw **170** is configured such that when it is screwed into the terminal block **130** at the set screw hole **131**, it presses on the power wire holding it in place within the wire receptacle **132**. As the set screw **170** is tightened into place against the power wire, the spring mechanism **174** is compressed. Should the power wire settle, the compressed spring mechanism **174** expands pushing the plunger **177** down along the set screw stem **175** and against the power wire maintaining pressure against the power wire. This pressure aids in holding the power wire in place within the wire receptacle **132** and also in maintaining a proper electrical contact between the set screw **170** and the power wire.

[0038] Computer system **100** may further include many components not specifically shown in FIG. **1** including storage and memory components, and networking components, for example.

[0039] Turning now to FIGS. **2-5B**, an example set screw **270** will be described. Reference will also be made to FIGS. **6A-7B** on occasion. The set screw **270** is one example implementation of the set screw **170** described above. Thus, set screw **270** comprises some components which correspond to (i.e., are implementations of) components of the set screw **170**. These corresponding components are assigned similar reference numbers having the same last two digits, such as **177** and **277**. Descriptions of the components of the set screw **170** above may apply to the corresponding components of the set screw **270**. Although set screw **270** is one example implementation of set screw **170**, set screw **170** is not limited to set screw **270**.

[0040] FIGS. **2-4** show different perspectives of example set screw **270**, while FIGS. **5A-B** show exploded views of the set screw. As shown in FIGS. **2** and **4**, the set screw includes a set screw

head 271, a spring mechanism 274, and a plunger 277. The set screw head 271 includes both the tool receptacle 272 and threading 273. The tool receptacle 272 is where an appropriate tool may be used to tighten and loosen the set screw 270 within the terminal block. In the illustrated example, the tool receptacle 272 has a hexagonal profile to receive a hex wrench, but any other shape of tool receptacle could be used instead, such as a Philips screw drive receptacle, flat-head screw driver receptacle, a square driver receptacle, a Torx driver receptacle, etc. The threading 273 is external threading which meshes with the internal threading of the terminal block as the set screw 270 is screwed into the terminal block. From the side-on perspective shown in FIG. 2, below the set screw head 271 is the spring mechanism 274 and below that is the plunger 277.

[0041] As shown in FIGS. 5A and 5B, the set screw 270 also comprises a set screw stem 275 and a retainer 276. The stem 275 extends distally from a bottom side of the set screw head 271. As shown in FIGS. 3, 5A and 5B (see also FIGS. 7A and 7B, described below), the retainer 276 is attached to a distal end portion of the stem 275 and the spring mechanism 274 and the plunger 277 are disposed on the stem 275 between the retainer 276 and the head 271. As shown in FIGS. 3, 5A, 5B, and 7A, the plunger 277 comprises a large plunger hole 279 that fits (has a larger diameter than) the retainer 276 and a smaller hole that fits (has a larger diameter than) the stem 275. The larger hole 279 and smaller hole 278 are coaxial and connected to one another, with the smaller hole 278 being proximal of the larger hole 279 (i.e., the smaller hole 278 is closer than the larger hole 279 to the screw head 271). The stem 275 extends through the smaller hole 278 into the larger hole 279. As shown in FIGS. 3 and 7A, the retainer 276 is positioned within the hole 279 and can move within the hole 279 as the plunger 277 moves along the stem 275. The smaller hole 278 has a diameter smaller than the retainer 267 such that the retainer 276 cannot be drawn through the hole 278. In other words, the retainer 276 abuts a portion of the plunger 277 which forms a distal rim of the smaller hole 278, which prevents the plunger 277 from moving distally past the retainer 276, thus retaining the plunger 277 on the stem 275. In this example, the retainer 276 is a clip which is removably attached to the stem 275 by engagement with a retainer holder 291 in the distal end portion of the stem 275, as shown in FIG. 5A. The retainer holder 291 may comprise a groove encircling the stem 275 and the retainer 267 may be snap-fitted into the groove to attach either retainer 276 to the stem 275.

[0042] As shown in FIGS. 5A and 7A, in some examples, the plunger 277 has a recess 292 at a proximal end thereof, i.e., the end where the plunger 277 meets the spring mechanism 274. The spring mechanism 277 may fit within the recess 292. In other examples, the recess 292 may be omitted (see FIGS. 8A and 8B, described below).

[0043] As shown in FIGS. 2, 5A, 5B, and 7B, in this example, the spring mechanism 274 comprises multiple conical disk springs 293. Specifically, in the illustrated example, eight disk springs 293 are arranged in the third configuration described above, i.e., in four pairs of springs, with each pair having two springs 293 with opposite orientations. However, any other configuration of the springs 293 could be used, and any other number of springs 293 could be used. For example, FIGS. 6A-6D illustrate some examples of configurations of spring mechanisms using conical disk springs which could be used to form the spring mechanism 274.

[0044] FIG. 6A illustrates an example configuration in which a single upright disk spring 500 is used. FIG. 6B illustrates another configuration in which multiple upright disk springs 500 may be stacked in series. This is one example of the second configuration described above. While three upright disk springs 500 are shown, it should be understood that any number of upright disk springs 500 are possible. In a series configuration such as in FIG. 6B, the spring forces of the disk springs 500 are added together such that a given amount of compression of the entire stack produces a total spring force which is multiplied relative to the spring force generated by compressing a single spring the same amount. Specifically, if a given disk spring 500 generates a spring force of X when compressed a given distance D , compression of the entire stack by that same amount D will generate a spring force of $N \cdot X$, where N is the number of disk springs 500.

[0045] FIG. 5C illustrates an example of the spring mechanism configured with pairs of disk springs, each comprising two springs **500** and **510** oriented in opposite directions. This is an example of the third configuration described above. Upright disk springs **500** are shown upright and each one is paired with an inverted disk spring **510**. Two such stacked pairs are illustrated, but in practice any number of pairs could be included. In such a configuration, the range of motion for the entire stack is extended as compared to if the same number of springs were provided in a series configuration. For example, if a single disk spring has a range of motion of D , then in the second configuration the range of motion of the entire stack is D whereas range of motion of the entire stack in the third configuration may be $D*N$, where N is the number of disks.

[0046] FIG. 5D shows another example configuration of the spring mechanism where multiple sets **512** of springs are stacked, with each set **512** of springs comprising two subsets **511**: a first subset **511a** comprising two upright disk springs **500** in series stacked on a second subset **511b** comprising two inverted disk springs **510** in series. This pattern is shown repeated twice in FIG. 5D (i.e., two stacked sets **512**), but it may be repeated many times with more sets **512**. Moreover, while only two disks **500** or **510** are shown per subset **511**, in practice each subset **511** could have more than two disks. This configuration may multiply the amount of spring force generated for a given amount of compression as compared to a single disk spring. Specifically, if M is the number of springs per subset and X is the spring force generated by compressing a single spring by a given distance D , then compressing the entire stack by the given distance D will produce a spring force of $M*D$. Moreover, this configuration may multiply the range of motion of the stack as compared to a purely series stack. For example, if a single disk spring has a range of motion of D , then the range of motion of the entire stack may be $Y*D$, where Y is the number of subsets **511** included in the stack.

[0047] While these four specific examples are illustrated in FIGS. 6A-6D, it should be understood that any number of disk springs in any pattern of upright and inverted configurations may be used.

[0048] Turning now to FIGS. 7A-8B, example electrical terminations **705** and **805** will be described. These electrical terminations **705** and **805** are example implementations of the electrical termination **105** described above. Electrical termination **705** comprises an instance of the set screw **270** described above, together with a terminal block **230**. Electrical termination **805** comprises a set screw **370**, together with the terminal block **230**. The set screw **870** is a variation of the set screw **270** and is identical thereto except that set screw **870** includes a plunger **877** in lieu of the plunger **277**, as will be described below. Components of the set screw **870** which are identical to those of the set screw **270** are given the same reference numbers, and duplicative descriptions thereof are omitted below.

[0049] FIGS. 7A-B show an example terminal block **230** and set screw **270** in a cross-section view. Terminal block **230** comprises a set screw hole **231** configured to receive the set screw **270**. The set screw hole **231** comprises internal terminal block threading **233** on a radially inward surface of the set screw hole **231** configured to mate with the external threading **273** of the set screw **270**. FIGS. 7A and 7B show the set screw head **271** screwed into the terminal block **230** where the threading **273** of the set screw head **271** has been mated with the terminal block threading **233**.

[0050] The terminal block **230** also comprises a wire receptacle **232** configured to receive a wire such as multistrand wire **280**. The set screw hole **231** extends along an axis which is transverse to an axis of the wire receptacle **232**, and screw hole **231** intersects with wire receptacle **232**. Thus, when set screw **270** is screwed into terminal block **230**, as in FIGS. 7A and 7B, a portion of set screw **270** is retained in set screw hole **231** (e.g., the head **271**) and a portion of set screw **270** protrudes out of set screw hole **231** into wire receptacle **232**. Specifically, when wire **280** is in receptacle **232** and set screw **270** is installed in the set screw hole **231** and appropriately tightened down, the plunger **277** of the set screw **270** will protrude into the wire receptacle **232** and contact the wire **280**, as shown in FIG. 7A.

[0051] In FIG. 7A, the set screw **270** is shown in a compressed state. In this state, plunger **277** is at a retracted position which is closer to the screw head **271** than a resting position of the plunger **277**.

(In the resting position, the plunger **277** is located as far distally along the stem **275** as possible, i.e., at the position where the rim around the smaller hole **278** of plunger **277** abuts the retainer **276**). With the plunger **277** in this retracted position, the spring mechanism **274** is in a loaded state where the disk springs are compressed, as shown in FIG. 7A.

[0052] The compressed state of the set screw **270** as shown in FIG. 7A may be attained, for example, in response to the set screw **270** being screwed into the terminal block **230** while the wire **280** is present. In other words, FIG. 7A depicts a state of the electrical termination **705** immediately after the set screw **270** is initially installed to hold the wire **280**. As the set screw **270** is screwed into the terminal block **230**, the plunger **277** eventually contacts the wire **280**. Continued tightening of the set screw **270** thereafter results in the set screw head **271** and stem **275** moving distally (downward in FIG. 7A) while the plunger **277** is held stationary by the wire **280**, resulting in the distance between plunger **277** and head **271** decreasing. In other words, plunger **277** moves relative to the stem **275** towards the head **271**. This results in the compression of the spring mechanism **274**. If tightening is continued, eventually the plunger **277** reaches the retracted position shown in FIG. 7A, with the spring mechanism **274** being fully compressed.

[0053] FIG. 7B shows the electrical termination **705** in a partially decompressed state. In this state, the plunger **277** is located at a position along the stem **275** which is part way between the fully retracted position shown in FIG. 7A and the resting, fully distal, position described above. This state may occur, for example, sometime after the state shown in FIG. 7A as a result of the wire **280** having settled. When the wire **280** settles, the spring forces generated by the spring mechanism **274**, which push the plunger **277** in the distal (downward) direction, cause the plunger **277** to follow the wire **280**. Thus, to reach the state of FIG. 7B, the plunger **277** will have moved distally (downward) a distance **701** relative to the retracted position of FIG. 7A. As a result of this movement, the plunger **277** is now farther from the set screw head **271**, and therefore the spring mechanism **274** is partially decompressed. However, the spring mechanism **274** remains partially compressed and thus continues to generate spring forces which push the plunger **277** against the wire **280**, thus maintaining the needed contact pressure to ensure a good electrical connection. This state of the spring mechanism **274** may also be referred to as a partially unloaded state. The distance **701** shows where the plunger **277** has moved further into the wire receptacle **232** pushed by the spring mechanism **274** as the wire **280** has settled. Settling of the wire **280** may happen over time and may be caused by fluctuations in heat and vibration.

[0054] As shown in FIGS. 8A-B, the electrical termination **805** is similar to the electrical termination **705** described above. In particular, electrical termination **805** comprises the same terminal block **230** described above. Moreover, electrical termination **805** comprises a set screw **870** which is identical to the set screw **270** except that the set screw **870** comprises the plunger **877** instead of the plunger **277**. The plunger **877** is similar to the plunger **277** except that the plunger **877** does not have the recess **292** at the proximal end thereof. Instead, the proximal end of the plunger **877** is flat. FIG. 8A illustrates the electrical termination **805** in the compressed state, which is similar to the compressed state of the electrical termination **705** described above and illustrated in FIG. 7A. FIG. 8B illustrates the electrical termination **805** in the partially compressed state, which is similar to the partially decompressed state of the electrical termination **705** described above and illustrated in FIG. 7B. In transitioning between these states, the plunger **877** moves distally (downward) by a distance **801**, as shown in FIG. 8B.

[0055] FIG. 9 shows an example terminal block unit **901**. This example shows three electrical terminations, each comprising a terminal block **930** and a corresponding set screw **970**. Each terminal block **930** comprises a set screw hole **931** and wire receptacle **932**. It should be understood that while this example terminal block unit **901** shows three terminal blocks **930**, any number could be used. Additionally, it is not required that there be an equal number of set screw holes **931** and wire receptacles **932**. For example, a terminal block could be configured where multiple set screws holes **931** are configured for a single wire receptacle **932**. Also shown in FIG. 9 are three set screws

970, each one screw into a set screw hole 931.

[0056] In the description above, various types of electronic circuitry are described. As used herein, “electronic” is intended to be understood broadly to include all types of circuitry utilizing electricity, including digital and analog circuitry, direct current (DC) and alternating current (AC) circuitry, and circuitry for converting electricity into another form of energy and circuitry for using electricity to perform other functions. In other words, as used herein there is no distinction between “electronic” circuitry and “electrical” circuitry.

[0057] It is to be understood that both the general description and the detailed description provide examples that are explanatory in nature and are intended to provide an understanding of the present disclosure without limiting the scope of the present disclosure. Various mechanical, compositional, structural, electronic, and operational changes may be made without departing from the scope of this description and the claims. In some instances, well-known circuits, structures, and techniques have not been shown or described in detail in order not to obscure the examples. Like numbers in two or more figures represent the same or similar elements.

[0058] In addition, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context indicates otherwise. Moreover, the terms “comprises”, “comprising”, “includes”, and the like specify the presence of stated features, steps, operations, elements, and/or components but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups. Components described as coupled may be electronically or mechanically directly coupled, or they may be indirectly coupled via one or more intermediate components, unless specifically noted otherwise. Mathematical and geometric terms are not necessarily intended to be used in accordance with their strict definitions unless the context of the description indicates otherwise, because a person having ordinary skill in the art would understand that, for example, a substantially similar element that functions in a substantially similar way could easily fall within the scope of a descriptive term even though the term also has a strict definition.

[0059] And/or: Occasionally the phrase “and/or” is used herein in conjunction with a list of items. This phrase means that any combination of items in the list—from a single item to all of the items and any permutation in between—may be included. Thus, for example, “A, B, and/or C” means “one of {A}, {B}, {C}, {A, B}, {A, C}, {C, B}, and {A, C, B}”

[0060] Elements and their associated aspects that are described in detail with reference to one example may, whenever practical, be included in other examples in which they are not specifically shown or described. For example, if an element is described in detail with reference to one example and is not described with reference to a second example, the element may nevertheless be claimed as included in the second example.

[0061] Unless otherwise noted herein or implied by the context, when terms of approximation such as “substantially,” “approximately,” “about,” “around,” “roughly,” and the like, are used, this should be understood as meaning that mathematical exactitude is not required and that instead a range of variation is being referred to that includes but is not strictly limited to the stated value, property, or relationship. In particular, in addition to any ranges explicitly stated herein (if any), the range of variation implied by the usage of such a term of approximation includes at least any inconsequential variations and also those variations that are typical in the relevant art for the type of item in question due to manufacturing or other tolerances. In any case, the range of variation may include at least values that are within $\pm 1\%$ of the stated value, property, or relationship unless indicated otherwise.

[0062] Further modifications and alternative examples will be apparent to those of ordinary skill in the art in view of the disclosure herein. For example, the devices and methods may include additional components or steps that were omitted from the diagrams and description for clarity of operation. Accordingly, this description is to be construed as illustrative only and is for the purpose of teaching those skilled in the art the general manner of carrying out the present teachings. It is to

be understood that the various examples shown and described herein are to be taken as exemplary. Elements and materials, and arrangements of those elements and materials, may be substituted for those illustrated and described herein, parts and processes may be reversed, and certain features of the present teachings may be utilized independently, all as would be apparent to one skilled in the art after having the benefit of the description herein. Changes may be made in the elements described herein without departing from the scope of the present teachings and following claims. [0063] It is to be understood that the particular examples set forth herein are non-limiting, and modifications to structure, dimensions, materials, and methodologies may be made without departing from the scope of the present teachings.

[0064] Other examples in accordance with the present disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with the following claims being entitled to their fullest breadth, including equivalents, under the applicable law.

Claims

1. A device, comprising: a terminal block comprising a set screw hole and a wire receptacle; a set screw, comprising: a set screw head configured to be received in the set screw hole, comprising: a tool receptacle; external threading around the outside of the set screw head configured to engage internal threading in the set screw hole; a set screw stem attached at a first end portion to the bottom of the set screw head; a plunger movably coupled to the set screw stem and movable along the set screw stem between an extended position and a retracted position; a retainer, connected at a second end portion of the set screw stem and configured to retain the plunger on the set screw stem; a spring mechanism, comprising one or more springs on the set screw stem between the set screw head and the plunger and configured to urge the plunger toward the extended position; wherein the wire receptacle is configured to receive a power wire, wherein in a state of the power wire received in the wire receptacle and the set screw installed in the set screw hole: the plunger extends into the wire receptacle and contacts the power wire; and the spring mechanism is compressed such that the spring mechanism urges the plunger against the power wire.
2. The device of claim 1, wherein the spring mechanism comprises a conical disk spring.
3. The device of claim 1, wherein the spring mechanism comprises a plurality of conical disk springs stacked in parallel.
4. The device of claim 1, wherein the spring mechanism comprises one or more pairs of conical disk springs stacked wherein each pair comprises two conical disk springs stacked in opposite directions from each other.
5. The device of claim 1, wherein upon completion of installation of the set screw, the plunger is located in a first position, and in response to the power wire experiencing settling subsequent to installation of the set screw, the spring mechanism causes the plunger to move to a second position and to maintain a predetermined contact pressure with the power wire, wherein the first position is closer to the retracted position than is the second position.
6. The device of claim 1, wherein the terminal block further comprises one or more additional set screw holes and wire receptacles.
7. The device of claim 1, wherein the end of the plunger that meets the spring mechanism comprises one of the following: a flat end and a recessed end.
8. A computing system, comprising: one or more compute trays each comprising a chassis, a system board supported by the chassis, and a processor mounted to the system board; a power distribution unit comprising: an input unit configured to receive power; an output unit configured to distribute the power to the compute trays; one or more terminal blocks configured to receive a power wire at a wire receptacle and connect the power wire to the power distribution unit, wherein in a state of the power wire connected to the power distribution unit, the power wire is configured

to convey the power to or through the power distribution unit, and wherein the one or more terminal blocks comprises a set screw, the set screw, comprising: a set screw head, comprising: a tool receptacle; external threading around the outside of the set screw head configured to engage internal threading in the terminal block; a set screw stem attached at a first end portion to the bottom of the set screw head; a plunger movably coupled to the set screw stem and movable along the set screw stem between an extended position and a retracted position; a retainer, connected at a second end portion of the set screw stem and configured to retain the plunger on the set screw stem; a spring mechanism, comprising one or more springs on the set screw stem between the set screw head and the plunger and configured to urge the plunger toward the extended position; wherein in a state of the power wire received in the wire receptacle and the set screw installed in the set screw hole: the plunger extends into the wire receptacle and contacts the power wire; and the spring mechanism is compressed such that the spring mechanism urges the plunger against the power wire.

9. The computing system of claim 8, wherein the spring mechanism comprises a conical disk spring.

10. The computing system of claim 8, wherein the spring mechanism comprises a plurality of conical disk springs stacked in parallel.

11. The computing system of claim 8, wherein the spring mechanism comprises one or more pairs of conical disk springs stacked wherein each pair comprises two conical disk springs stacked in opposite directions from each other.

12. The computing system of claim 8, wherein upon completion of installation of the set screw, the plunger is located in a first position, and in response to the power wire experiencing settling subsequent to installation of the set screw, the spring mechanism causes the plunger to move to a second position and to maintain a predetermined contact pressure with the power wire, wherein the first position is closer to the retracted position than is the second position.

13. The computing system of claim 8, wherein the one or more terminal blocks further comprise one or more additional set screw holes and wire receptacles.

14. The computing system of claim 8, wherein the power distribution unit comprises one or more circuit breaker mechanisms, and wherein the one or more circuit breaker mechanisms comprise one or more terminal blocks.

15. The computing system of claim 8, wherein the input unit comprises one or more terminal blocks.

16. The computing system of claim 8, wherein the power distribution unit comprises one or more terminal blocks disposed between the input unit and the output unit.

17. The computing system of claim 8, wherein the one or more compute trays are housed in a rack system.

18. The computing system of claim 8, wherein the one or more compute trays comprise a power supply unit configured to accept power from the output unit of the power distribution unit and distribute it to components of the compute tray.

19. The computing system of claim 18, wherein the power supply unit comprises one or more terminal blocks.

20. The computing system of claim 8, wherein the end of the plunger that meets the spring mechanism comprises one of the following: a flat end and a recessed end.
